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**Real-Time Multi-Modal Image Manipulation Detection Using Cross-Modal
Transformer Fusion: A Multi-Stream Forensic Approach and Cross-Dataset
Generalisation Analysis**

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Certified that the work contained in the thesis titled “*Real-Time Multi-Modal Image Manipulation Detection Using Cross-Modal Transformer Fusion: A Multi-Stream Forensic Approach and Cross-Dataset Generalisation Analysis*”, by *Subhankar Roy*, 2024DS19 has been carried out under my supervision and this work has not been submitted elsewhere for a degree.

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सिद्धिर्भवति कर्मजा

– *Dedicated to my parents and all the mentors whose wisdom and encouragement
light my path.*

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- Subhankar Roy

Abstract

Digital image manipulation has become increasingly sophisticated with the advent of advanced editing tools, posing significant threats to the authenticity of photographic content across journalism, legal proceedings, and social media. Automated image manipulation detection has thus emerged as a critical problem in the field of multimedia forensics. This thesis presents a progressive ablation study of five architectures for real-time multi-modal image manipulation detection, progressing from traditional forensic baselines to a novel deep learning architecture.

The proposed system, *CrossModalFusionNet*, integrates five complementary forensic streams—RGB, Error Level Analysis (ELA), Noise Residual, DCT Log-Magnitude, and Vision Transformer (ViT) features—fused through a cross-modal transformer encoder that models inter-modality dependencies. Rather than assigning static or scalar attention weights, the transformer encoder enables each forensic stream to attend to and contextualise all other streams, learning richer manipulation-discriminative representations.

All models are trained on the CASIA2 dataset (12,617 images) and evaluated on a held-out CASIA2 test set as well as two out-of-distribution datasets—RAISE (363 images) and IMD2020 (2,417 images)—without any fine-tuning, to assess cross-dataset generalisation. *CrossModalFusionNet* achieves an F1 score of **0.9446** and AUC-ROC of **0.9927** on CASIA2, representing a **7.85%** improvement in F1 over the best prior baseline (ViT Fusion, $F1 = 0.8786$). On out-of-distribution datasets, *CrossModalFusionNet* achieves the best absolute F1 on both RAISE (0.7277) and IMD2020 (0.7276), though with a larger generalisation gap than simpler architectures, highlighting a complexity–generalisation trade-off. Inference latency profiling confirms that all models operate within real-time GPU thresholds (*CrossModalFusionNet*: 18.00 ms/image), demonstrating the practical deployability of the proposed system.

The thesis further extends *CrossModalFusionNet* to a three-class detector—Authentic, Traditionally Manipulated, and AI-Generated—through partial fine-tuning on CASIA2 and CIFAKE (Stable Diffusion v1.4). The 3-class model achieves **96.36%** accuracy and a macro F1 of **0.9638**, with perfect detection of AI-generated images ($F1 = 1.000$), demonstrating the framework’s extensibility to the emerging threat of generative AI-produced content.

सारांश

डिजिटल इमेज में हेरफेर उन्नत संपादन उपकरणों के प्रसार के साथ अत्यंत परिष्कृत हो गया है, जिससे पत्रकारिता, कानूनी कार्यवाही और सोशल मीडिया में फोटोग्राफिक सामग्री की प्रामाणिकता को गम्भीर खतरा उत्पन्न हो रहा है। स्वचालित इमेज मैनिपुलेशन डिटेक्शन मल्टीमीडिया फॉरेंसिक के क्षेत्र में एक महत्वपूर्ण समस्या बन गई है। यह शोध प्रबंध रियल-टाइम मल्टी-मोडल इमेज मैनिपुलेशन डिटेक्शन हेतु पाँच आर्किटेक्चर का प्रगतिशील एब्लेशन अध्ययन प्रस्तुत करता है, जो पारम्परिक फॉरेंसिक बेसलाइन से एक नवीन डीप लर्निंग आर्किटेक्चर की ओर अग्रसर होता है।

प्रस्तावित प्रणाली CrossModalFusionNet पाँच पूरक फॉरेंसिक स्ट्रीम—RGB, एरर लेवल एनालिसिस (ELA), नॉइज़ रेजिड्यूअल, DCT लॉग-मैग्निट्यूड, तथा विज़न ट्रांसफॉर्मर (ViT) फीचर्स—को एकीकृत करती है। ये पाँचों स्ट्रीम एक क्रॉस-मोडल ट्रांसफॉर्मर एन्कोडर के माध्यम से फ्यूज़ किए जाते हैं, जो इंटर-मोडैलिटी निर्भरताओं का मॉडलन करता है। स्थैतिक अदिश अटेंशन भार के स्थान पर, ट्रांसफॉर्मर एन्कोडर प्रत्येक फॉरेंसिक स्ट्रीम को अन्य सभी स्ट्रीम पर ध्यान देने तथा उनसे संदर्भ ग्रहण करने में सक्षम बनाता है।

सभी मॉडलों को CASIA2 डेटासेट (12,617 इमेज) पर प्रशिक्षित किया गया है और CASIA2 टेस्ट सेट के साथ-साथ दो आउट-ऑफ-डिस्ट्रिब्यूशन डेटासेट—RAISE (363 इमेज) और IMD2020 (2,417 इमेज)—पर बिना किसी फाइन-ट्यूनिंग के परीक्षण किया गया है। CrossModalFusionNet ने CASIA2 पर F1 स्कोर **0.9446** और AUC-ROC **0.9927** प्राप्त किया, जो सर्वश्रेष्ठ पूर्व बेसलाइन (ViT Fusion, $F1 = 0.8786$) की तुलना में F1 में **7.85%** का सुधार दर्शाता है। आउट-ऑफ-डिस्ट्रिब्यूशन डेटासेट पर CrossModalFusionNet ने RAISE (0.7277) और IMD2020 (0.7276) दोनों पर सर्वश्रेष्ठ निरपेक्ष F1 प्राप्त किया। इन्फेरेंस लेटेंसी प्रोफाइलिंग यह पुष्टि करती है कि सभी मॉडल रियल-टाइम GPU थ्रेशोल्ड के भीतर कार्य करते हैं (CrossModalFusionNet: 18.00 ms/इमेज)।

यह शोध प्रबंध CrossModalFusionNet को आगे तीन-वर्ग डिटेक्टर—प्रामाणिक (Authentic), पारम्परिक रूप से हेरफेर किए गए (Traditionally Manipulated), और AI-जनित (AI-Generated)—तक विस्तारित करता है। 3-वर्ग मॉडल ने **96.36%** सटीकता और मैक्रो F1 **0.9638** प्राप्त किया, जिसमें AI-जनित इमेज की पूर्ण डिटेक्शन ($F1 = 1.000$) शामिल है। यह इस ढाँचे की विस्तारणीयता को जनरेटिव AI द्वारा उत्पन्न सामग्री के उभरते खतरे की दिशा में सिद्ध करती है।

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Chapter 1

Introduction

1.1 Background and Motivation

The integrity of digital images has become a cornerstone of trust in modern information systems. From forensic evidence and photojournalism to medical imaging and social media, the authenticity of photographic content directly impacts decision-making in critical domains. However, the rapid proliferation of accessible and powerful image editing tools—ranging from open-source software to AI-based generative models—has made it increasingly trivial to create convincing manipulated images that are imperceptible to the human eye.

Image manipulation encompasses a broad range of operations: *splicing* (inserting a region from one image into another), *copy-move* (duplicating a region within the same image), and *removal* (erasing objects and inpainting the background). Each operation leaves subtle statistical footprints in the image that are not visible to human observers but are detectable through forensic analysis.

Traditional image forensics relied on hand-crafted statistical features derived from physical properties of images. Error Level Analysis (ELA) [1], which detects inconsistencies in JPEG re-compression residuals, is one of the most established such techniques. Similarly, noise residual analysis exploits the fact that different regions of a manipulated image may originate from different cameras or imaging conditions, leading to spatially inconsistent noise patterns. Despite their theoretical grounding, hand-crafted approaches

exhibit limited accuracy on modern high-quality manipulations and are sensitive to post-processing operations such as re-compression and resizing.

Deep learning has transformed image recognition and classification, and its application to image forensics has yielded substantial accuracy gains over traditional methods. Convolutional Neural Networks (CNNs) can learn hierarchical feature representations directly from raw pixel data, capturing manipulation artifacts at multiple scales [2]. More recently, Vision Transformers (ViT) [3] have demonstrated that global context captured through self-attention mechanisms provides complementary information to the local convolutional features, making them attractive for detecting globally inconsistent manipulations.

A key insight from recent forensic research is that different modalities of the same image carry complementary forensic information. The RGB image reflects visual content, noise residuals reveal high-frequency sensor-level inconsistencies, DCT log-magnitude spectra expose frequency-domain blocking artifacts, and ELA reveals compression-level anomalies. No single modality captures all aspects of manipulation evidence; fusing multiple modalities can thus yield more robust detection.

1.2 Problem Statement

Given a digital image I , the binary classification task is:

$$f(I) = \begin{cases} 0 & \text{if } I \text{ is authentic} \\ 1 & \text{if } I \text{ has been manipulated} \end{cases}$$

The challenges addressed in this thesis are:

1. **Multi-modal feature fusion:** How to effectively combine multiple forensic streams (RGB, ELA, noise, DCT, ViT) such that each stream contextualises the others.
2. **Scalability and real-time deployability:** The model must process high-resolution images within real-time latency thresholds (<33 ms per image at 30 fps).
3. **Cross-dataset generalisation:** Models trained on one dataset must generalise to

unseen datasets with different camera types, compression pipelines, and manipulation methods.

1.3 Contributions

This thesis makes the following original contributions:

- C1.** A systematic progressive ablation study across five architectures, from a traditional ELA + Random Forest baseline to a novel cross-modal transformer fusion network, providing a clear performance trajectory on three public benchmarks.
- C2. CrossModalFusionNet:** a novel five-stream architecture that fuses RGB, ELA (as a learned CNN stream), Noise Residual, DCT Log-Magnitude, and ViT features using a cross-modal transformer encoder. The transformer is applied over the *modality dimension*, enabling each forensic stream to attend to and contextualise all others.
- C3.** Empirical demonstration that cross-modal transformer fusion achieves $F1 = 0.9446$ and $AUC = 0.9927$ on CASIA2, a **+7.85% F1** improvement over the best prior baseline.
- C4.** Cross-dataset generalisation analysis on RAISE and IMD2020 without fine-tuning, providing a rigorous assessment of which architectural choices promote dataset-agnostic forensic feature learning.

1.4 Scope and Research Context

The proliferation of generative AI tools has created a new category of visual content that existing forensic systems were not designed to handle. Traditional image manipulation detection treats authenticity as a binary property: an image is either an unmodified photograph or a composited forgery created by splicing, copy-move, or inpainting operations. However, content synthesised by diffusion models such as Stable Diffusion, or by Generative Adversarial Networks (GANs) [4], represents a qualitatively different

threat — it is neither a real photograph nor a traditional composite. It is a fully synthesised image with a fundamentally distinct statistical fingerprint that no conventional forensic stream was designed to identify.

This distinction has concrete consequences for deployed forensic systems. A model trained exclusively on CASIA2 — which contains only splicing and copy-move manipulations — will encounter Stable Diffusion images as a novel, out-of-distribution input class. As demonstrated in Section 5.7, the binary CrossModalFusionNet systematically classifies such images as *authentic*. This failure mode is not random; it is causally explainable: synthesised images have never undergone JPEG compression, sensor noise processing, or region compositing. Every forensic stream — ELA, Noise Residual, and DCT — returns a “clean” reading, indistinguishable from an unmodified photograph in the feature space the model was trained on.

The scope of this thesis therefore encompasses two interconnected problems:

1. **Binary manipulation detection:** distinguishing authentic photographs from traditionally manipulated images (splicing, copy-move, inpainting) using a progressive multi-modal deep learning framework evaluated on the CASIA2 benchmark and two zero-shot out-of-distribution datasets (RAISE, IMD2020).
2. **Ternary content classification:** extending binary detection to a three-class framework that additionally identifies AI-generated content through parameter-efficient partial fine-tuning on CASIA2 and CIFAKE [5].

This work does not address video forensics, pixel-level manipulation localisation, or GAN fingerprint attribution — each of which constitutes a distinct and active research direction. These are identified as natural extensions in Chapter 6.

1.5 Thesis Organisation

The remainder of this thesis is organised as follows:

- **Chapter 2** reviews related work in traditional image forensics, deep learning for forgery detection, and transformer-based approaches.

- **Chapter 3** describes the datasets used and the forensic feature extraction and pre-processing pipeline.
- **Chapter 4** presents all five model architectures in the ablation study, with particular detail on the proposed CrossModalFusionNet.
- **Chapter 5** describes the experimental setup and presents quantitative results, ablation analysis, and inference latency profiling.
- **Chapter 6** concludes the thesis with a summary of findings and directions for future work.

Chapter 2

Literature Review

2.1 Traditional Image Forensics

Early image forensic research focused on physically motivated hand-crafted features derived from the properties of image acquisition and compression pipelines.

2.1.1 Error Level Analysis

Krawetz [1] introduced Error Level Analysis (ELA) to detect JPEG re-saving artifacts. The core insight is that uniform JPEG compression leaves predictable error patterns, while spliced regions—which may have been compressed at a different quality—produce anomalous residuals. Despite its simplicity, ELA remains a widely used forensic tool and forms the traditional baseline in this work.

2.1.2 Re-sampling and Interpolation Detection

Popescu and Farid [6] demonstrated that copy-move forgeries often introduce periodic correlations in the image signal due to geometric transformations (rotation, scaling) applied before pasting. Their EM-based algorithm detects these periodic artifacts in the frequency domain, providing strong evidence for copy-move manipulation.

2.1.3 Noise Pattern Analysis

Fridrich et al. [7] proposed the Rich Model (SRM) for steganalysis, extracting high-dimensional noise residuals using 30 diverse filter bases. The same principle has been applied to forgery detection: manipulated regions generally exhibit noise characteristics inconsistent with the host image. This motivates the noise residual stream used in our multi-modal architectures.

2.1.4 DCT-based Approaches

Discrete Cosine Transform (DCT) coefficient analysis [8] exploits the fact that JPEG compression imposes a periodic block structure in the DCT domain. Inconsistencies in DCT statistics between authentic and spliced regions are detectable through spectral analysis, forming the basis of the DCT log-magnitude forensic stream. A comprehensive survey of JPEG forensic methods, covering both classical quantisation-table analysis and modern deep-learning variants, is provided by Wang et al. [9].

2.2 Deep Learning for Forgery Detection

2.2.1 Constrained CNNs

Bayar and Stamm [10] introduced constrained convolutional layers that are explicitly trained to suppress image content and amplify manipulation traces. Unlike standard CNNs that may learn content features, constrained CNNs are biased toward learning manipulation-discriminative residuals. This remains a strong baseline for forgery detection.

2.2.2 Multi-Scale and Multi-Branch Architectures

Rao and Ni [11] proposed a two-stream approach combining RGB features and SRM noise features. Zhou et al. [12] extended this to a two-branch architecture with cross-attention between the RGB and noise branches, showing that cross-modal interactions improve detection accuracy compared to independent stream processing.

2.2.3 ManTra-Net

Wu et al. [13] trained ManTra-Net on 385 different manipulation types, learning an anomaly feature representation that achieves both detection and localisation. ManTra-Net demonstrated that training on diverse manipulation types improves generalisation, a principle that motivates the multi-modal approach in this work.

2.2.4 MVSS-Net

Dong et al. [14] proposed Multi-View Self-Supervised (MVSS-Net), which exploits edge-supervised features and dual-stream self-supervised learning. MVSS-Net achieves state-of-the-art performance on several benchmarks and highlights the importance of edge consistency as a forensic cue. Building on this direction, Zhang et al. [15] proposed MEN-IML, a multi-scale edge-aware network that jointly performs manipulation detection and localisation by progressively fusing edge cues across spatial scales, achieving improved localisation precision over prior dual-branch approaches.

2.2.5 PSCC-Net and Progressive Multi-Scale Approaches

Liu et al. [16] proposed PSCC-Net, a Progressive Spatio-Channel Correlation Network for image manipulation detection and localisation. The architecture propagates detection confidence progressively from low-resolution to high-resolution feature maps through a top-down decoder, enabling localisation at pixel level. Unlike our work, PSCC-Net targets both detection and localisation jointly but does not incorporate multiple heterogeneous forensic modalities (ELA, DCT, noise residual) as independent learned streams. Its spatio-channel correlation design is orthogonal to our cross-modal fusion strategy and represents a promising direction for future integration, particularly for extending CrossModalFusionNet to produce pixel-level manipulation masks.

2.3 Vision Transformers in Forensics

2.3.1 Vision Transformer (ViT)

Dosovitskiy et al. [3] demonstrated that a pure transformer applied to image patches—the Vision Transformer (ViT-B/16)—achieves competitive image recognition performance when pre-trained on large datasets (ImageNet-21k, JFT-300M). The global self-attention mechanism allows ViT to capture long-range spatial dependencies that are difficult for local convolutional operations, making it sensitive to globally inconsistent image regions.

2.3.2 ViT-Based Forgery Detection

Recent work [17] has shown that ViT attention maps correlate with tampered regions, suggesting that global attention naturally identifies visually inconsistent patches. TruFor [18] combines CNN-based noise features with transformer-based RGB analysis, achieving strong performance on multiple public benchmarks. Our work differs by explicitly projecting ViT features into the same embedding space as EfficientNet-B0 features and fusing them with other forensic streams through cross-modal transformer attention. A broader treatment of manipulation detection, including face-specific deepfake methods, is provided in the Handbook of Digital Face Manipulation and Detection [19].

2.4 Multi-Modal Fusion Strategies

Fusion of multiple forensic modalities can be performed at different stages:

- **Early fusion** (input concatenation): modalities are concatenated at the pixel level before processing. Simple but loses modality-specific representations.
- **Late fusion** (decision aggregation): each modality produces an independent prediction, which are then aggregated. Robust but ignores inter-modality correlations.

- **Feature-level fusion:** modality features are combined at an intermediate representation level, allowing inter-modality interaction while preserving modality-specific processing.

Among feature-level strategies, attention-based fusion [20] has become dominant. Scalar softmax attention assigns a single weight per modality, while our proposed cross-modal transformer assigns contextualised, token-level attention across the full feature dimension, capturing richer inter-modality dependencies.

2.5 Summary and Research Gap

While existing multi-modal forensic methods achieve strong in-distribution performance, several gaps remain:

1. **Scalar vs. cross-modal attention:** Most works use simple scalar weights or concatenation for modality fusion, losing inter-modality contextual information.
2. **ELA as a learned stream:** ELA has traditionally been used only as a source of hand-crafted statistical features; its potential as a learned visual stream processed by a CNN backbone has not been explored.
3. **Comprehensive generalisation evaluation:** Many published systems are evaluated only on their training dataset split, with limited cross-dataset analysis.

This thesis directly addresses these gaps through CrossModalFusionNet and a rigorous three-dataset evaluation protocol.

2.6 Generative AI Content Detection

The rapid advancement of deep generative models—Generative Adversarial Networks (GANs) [4], Variational Autoencoders, and diffusion models—has introduced a qualitatively new challenge for image forensics. Unlike traditional manipulation, which composites or modifies existing photographs, generative AI synthesises photorealistic images from noise, producing content that has never existed as a physical photograph.

2.6.1 GAN Fingerprints and Spectral Artefacts

Wang et al. [21] demonstrated that CNN-generated images carry detectable spectral fingerprints in the frequency domain. A linear classifier trained on frequency-domain features generalised to unseen GAN architectures with surprisingly high accuracy, indicating that generated images share common spectral characteristics arising from the up-sampling operations in generator networks. These artefacts manifest as periodic peaks in the DCT log-magnitude spectrum at frequencies corresponding to generator stride patterns — a finding that directly motivates the inclusion of DCT log-magnitude as an independent forensic stream in CrossModalFusionNet.

2.6.2 Diffusion Model Detection

Diffusion models (Stable Diffusion, DALL-E, Midjourney) have largely superseded GANs in terms of output quality, producing images whose photorealism makes visual detection near-impossible. Corvi et al. [22] evaluated the transferability of GAN-era detectors to diffusion model outputs, finding significant performance degradation when models trained on GAN-generated content are applied to diffusion-generated content. They proposed spectral analysis-based detectors tailored to the high-frequency properties of diffusion-generated content.

Our empirical observation (Section 5.7) is consistent with this literature: Stable Diffusion images exhibit near-zero ELA residuals and spatially uniform noise floors because the generation pipeline does not involve JPEG re-compression or camera sensor processing. The *absence* of forensic signals — rather than the presence of detectable artefacts — becomes the distinguishing feature, and the cross-modal transformer learns to identify this distinctive silence across all five forensic streams jointly.

2.6.3 CIFAKE Dataset

Bird and Lotfi [5] introduced the CIFAKE dataset, containing 60,000 real images (sourced from CIFAR-10 [23]) and 60,000 AI-generated counterparts produced by Stable Diffusion v1.4 conditioned on CIFAR-10 class labels. CIFAKE provides a large-scale resource for binary real/AI-generated classification tasks. In this work, 5,123 AI-generated

images from CIFAKE are used as the third class in the 3-class partial fine-tuning protocol described in Section 5.7.

2.7 Comparative Summary of Related Work

Table 1 summarises the key prior methods compared to CrossModalFusionNet across the dimensions most relevant to this thesis: use of ELA, number of forensic streams, cross-modal attention, rigorous out-of-distribution evaluation, and AI-generated content detection capability.

Table 1: Comparative summary of image forensic methods

Method	Venue	ELA	Streams	X-Modal Attn	OOD Eval	AI-Gen
ELA + RF [1]	—	✓	—	×	×	×
Constrained CNN [10]	TIFS 2018	×	1	×	×	×
Rao & Ni [11]	WIFS 2016	×	2	×	×	×
Zhou et al. [12]	CVPR 2018	×	2	Partial	×	×
ManTra-Net [13]	CVPR 2019	×	1	×	✓	×
MVSS-Net [14]	TPAMI 2022	×	2	×	✓	×
PSCC-Net [16]	TCSVT 2022	×	Multi	×	✓	×
TruFor [18]	CVPR 2023	×	2	✓	✓	×
CrossModalFusionNet (Ours)	—	✓	5	✓	✓	✓

CrossModalFusionNet is the only method in this comparison that combines all five capabilities: ELA as a learned visual stream, five forensic modalities, full cross-modal attention, out-of-distribution evaluation, and AI-generated content detection. TruFor is

the most similar prior work in terms of architecture philosophy (ViT-based with a noise stream and transformer fusion), but differs in stream count, fusion depth, and detection scope.

Chapter 3

Datasets and Preprocessing

3.1 Datasets

Three publicly available datasets are used in this work. CASIA2 serves as the primary training and in-distribution evaluation dataset, while RAISE and IMD2020 are reserved exclusively for cross-dataset generalisation evaluation.

3.1.1 CASIA2

The CASIA Image Tampering Detection Evaluation Database v2.0 [24] is one of the most widely used benchmarks in image forensics. It contains 7,492 authentic images and 5,123 tampered images, spanning diverse categories including animals, architecture, nature, people, and objects. Three types of manipulation are represented:

- **Splicing:** regions from one image are inserted into another.
- **Copy-move:** a region within the same image is duplicated to another location.
- **Removal:** an object is removed and the background is inpainted.

The dataset is split into train/validation/test partitions at a 70/15/15 ratio using a fixed random seed (42) to ensure reproducibility.

3.1.2 RAISE

The RAISE dataset [25] contains 185 authentic raw images captured by four different digital cameras, and 178 corresponding spliced images. RAISE differs from CASIA2 in that images are captured in RAW format (then converted to TIFF/JPEG), offering a different noise profile and dynamic range. The manipulation pipeline also differs from CASIA2, making RAISE a strong test of cross-dataset generalisation.

3.1.3 IMD2020

IMD2020 [26] is a large-scale real-world dataset containing 2,010 manipulated images collected from the internet, with 414 authentic counterparts sourced from news agencies. The dataset is heavily class-imbalanced (approximately 5:1 manipulated-to-authentic) and includes diverse manipulation qualities ranging from crude composites to highly realistic forgeries, making it the most challenging evaluation dataset in this study.

3.1.4 CIFAKE: AI-Generated Image Dataset

CIFAKE [5] provides 60,000 real images and 60,000 AI-generated counterparts. The real images are sourced from CIFAR-10 [23], while the synthetic images are produced by Stable Diffusion v1.4 conditioned on CIFAR-10 class labels. All images are 32×32 pixels in the original dataset; they are upsampled to 224×224 using bilinear interpolation for compatibility with EfficientNet-B0 and ViT-B/16 input requirements.

CIFAKE is used exclusively for the 3-class fine-tuning phase and is not included in any 2-class training or evaluation. A key forensic property of CIFAKE images is relevant to this work: as outputs of a latent diffusion generation pipeline, they have never undergone JPEG compression, camera sensor processing, or image compositing. Consequently:

- **ELA residuals are near-zero:** no JPEG re-compression difference to measure.
- **Noise residuals are spatially uniform:** no camera sensor noise heterogeneity.
- **DCT spectra are clean:** no blocking artefacts or compositing discontinuities.

This distinctive three-stream “silence” is qualitatively different from both authentic photographs and traditionally manipulated images, enabling the partial fine-tuned model to achieve perfect AI-generated detection ($F1 = 1.000$) as reported in Section 5.7.

Table 2: Dataset statistics used in this work

Dataset	Authentic	Manipulated	Total	Role
CASIA2	7,492	5,125	12,617	Train / Val / Test
RAISE	185	178	363	Cross-dataset eval
IMD2020	414	2,003	2,417	Cross-dataset eval

Training class distribution: 5,231 authentic vs. 3,598 manipulated (59.2% / 40.8%).

3.2 Forensic Modality Extraction

Four forensic modalities are extracted from each image, each capturing complementary manipulation evidence.

3.2.1 Error Level Analysis (ELA)

ELA computes per-pixel JPEG re-compression residuals to detect regions saved at different compression levels:

$$ELA(I) = |I - \text{JPEG}_{q=95}(I)| \quad (1)$$

For the Random Forest baseline, 12 statistical features are derived: mean, standard deviation, maximum, minimum, 25th/50th/75th percentiles, quadrant spatial variance, edge-to-non-edge error ratio, and a 5-bin error histogram. For deep learning models, ELA is treated as a visual modality: the grayscale residual is normalised to $[0, 255]$ and converted to a 3-channel image for CNN input.

3.2.2 Noise Residual

The noise residual isolates high-frequency content by subtracting a Gaussian-blurred version of the image:

$$N(I) = I - \mathcal{G}_{5 \times 5}(I) \quad (2)$$

where $\mathcal{G}_{5 \times 5}$ denotes Gaussian blurring with a 5×5 kernel. This amplifies sensor noise and high-frequency manipulation traces while suppressing smooth content regions. Spliced regions originating from different cameras typically exhibit inconsistent noise patterns detectable through this modality.

3.2.3 DCT Log-Magnitude

The DCT log-magnitude modality exposes frequency-domain blocking and ringing artifacts introduced by JPEG compression and image compositing:

$$D(I) = \text{Norm}(\log(|\text{DCT}(\text{Gray}(I))| + \epsilon)) \quad (3)$$

where $\epsilon = 10^{-6}$ prevents numerical instability. The result is normalised to $[0, 255]$ and converted to a 3-channel image. Copy-paste operations between images compressed at different quality levels produce detectable DCT coefficient inconsistencies.

3.2.4 RGB Image

The original RGB image is retained as a modality to provide content-level features that contextualise the forensic signals from ELA, noise residual, and DCT streams.

3.3 Data Preprocessing and Augmentation

3.3.1 Training Augmentation

Training images undergo the following augmentation pipeline:

1. Resize to 256×256
2. Random resized crop to 224×224 (scale factor: $[0.9, 1.0]$)

3. Colour jitter (brightness ± 0.2 , contrast ± 0.2 , saturation ± 0.2 , hue ± 0.1)
4. Random horizontal flip (probability: 0.5)
5. Random rotation ($\pm 5^\circ$)
6. Normalisation: mean $\mu = [0.485, 0.456, 0.406]$, std $\sigma = [0.229, 0.224, 0.225]$

3.3.2 Validation and Test Preprocessing

Validation and test images (including cross-dataset evaluation) are only resized to 224×224 and normalised with the same ImageNet statistics. No augmentation is applied.

3.3.3 Dataset Classes and Loaders

Three dataset classes are implemented:

- `ForensicsDataset`: returns (rgb, label) for the single-stream CNN baseline.
- `MultiModalDataset`: returns (rgb, noise, dct, label) for 3- and 4-stream models.
- `MultiModal5Dataset`: returns (rgb, ela, noise, dct, label) for the 5-stream CrossModalFusionNet.

All forensic modalities are computed on-the-fly in the dataset `__getitem__` method, with `num_workers=0` to ensure notebook-defined functions remain in scope during `DataLoader` iteration.

3.4 Class Imbalance Analysis and Handling Strategy

3.4.1 CASIA2 Imbalance

CASIA2 exhibits a moderate class imbalance: 7,492 authentic images versus 5,123 manipulated images (ratio $\approx 1.46:1$). Within the training partition (70%), this yields 5,231 authentic versus 3,598 manipulated samples. Without correction, this imbalance would

bias the classifier toward predicting “authentic,” inflating accuracy while depressing recall on the manipulated class—which is more forensically costly.

Class-balanced cross-entropy loss is applied during training to counteract this effect:

$$\mathcal{L}_{\text{balanced}} = - \sum_{c \in \{0,1\}} \omega_c \log p(y = c \mid x) \quad (4)$$

where $\omega_c = N / (K \cdot n_c)$, N is the total number of training samples, $K = 2$ is the number of classes, and n_c the count of class c in the training set. For CASIA2: $\omega_{\text{auth}} \approx 0.84$ and $\omega_{\text{manip}} \approx 1.22$, up-weighting manipulated samples proportionally.

3.4.2 IMD2020 Class Imbalance

IMD2020 presents a severe 5:1 imbalance (2,010 manipulated versus 414 authentic). As an exclusively held-out cross-dataset evaluation set, no re-balancing is applied — the evaluation reflects the natural distribution of the dataset. This is important: applying over- or under-sampling to the test set would yield misleadingly optimistic metrics that do not reflect real-world deployment conditions.

The practical consequence is that *accuracy is a misleading metric for IMD2020*: a trivial all-manipulated classifier would achieve $\sim 83\%$ accuracy. For this reason, F1 score — which penalises poor recall on the minority class proportional to its support — is the primary reported metric for IMD2020 throughout this thesis.

3.4.3 3-Class Dataset Balancing

For the 3-class fine-tuning phase, classes are explicitly balanced at 5,123 images per class, limited by the size of the CASIA2 Manipulated partition. This produces a perfectly balanced 15,369-image corpus (Table 16) and eliminates any need for loss re-weighting in the fine-tuning stage.

Table 3: Summary of class imbalance across all datasets used

Dataset	Role	Class 0 (Auth)	Class 1 (Manip)	Ratio
CASIA2 (train split)	Training	5,231	3,598	1.45:1
CASIA2 (test split)	In-dist. eval	1,127	766	1.47:1
RAISE	OOD eval	185	178	1.04:1
IMD2020	OOD eval	414	2,003	1:4.84
3-class fine-tune	Fine-tuning	5,123	5,123	1:1 (balanced)

Chapter 4

System Architecture

This chapter presents all five architectures in the progressive ablation study. Each architecture builds upon the insights from the previous, culminating in the proposed Cross-ModalFusionNet.

4.1 Section 6: Traditional Baseline — ELA + Random Forest

The traditional baseline uses 12-dimensional ELA statistical features (Section 3) extracted from each image. A Random Forest classifier with 100 estimators and maximum depth 10 is trained on an 80/20 train/test split of CASIA2. This baseline represents the performance achievable with physics-based forensic features and a conventional machine learning classifier, without any end-to-end learned representations.

4.2 Section 7: Single-Stream CNN (EfficientNet-B0)

The single-stream CNN establishes the deep learning baseline. An EfficientNet-B0 [27] backbone, pre-trained on ImageNet, processes the RGB image. The final classification head is replaced with a linear layer mapping to 2 classes:

$$y = W_c \cdot \text{EfficientNet-B0}(I_{\text{RGB}})$$

This model contains no forensic priors; all manipulation-discriminative features are learned from RGB data alone.

4.3 Section 8: Multi-Stream CNN with Channel Gating

The first multi-modal model uses three parallel EfficientNet-B0 backbones to independently process RGB, Noise Residual, and DCT modalities. Per-stream sigmoid gates modulate each feature vector before concatenation:

$$\hat{f}_s = f_s \odot \sigma(W_s f_s + b_s), \quad s \in \{\text{RGB, Noise, DCT}\} \quad (5)$$

$$y = \text{FC}_{512 \rightarrow 2}(\text{ReLU}(\text{FC}_{3840 \rightarrow 512}([\hat{f}_{\text{RGB}}; \hat{f}_{\text{Noise}}; \hat{f}_{\text{DCT}}]))) \quad (6)$$

where $f_s \in \mathbb{R}^{1280}$ is the EfficientNet-B0 feature vector for stream s and σ is the sigmoid function. The gating mechanism allows the model to suppress uninformative channels on a per-feature basis without discarding entire streams.

4.4 Section 9: AdaptiveFusionNet — Learnable Softmax Attention

AdaptiveFusionNet replaces fixed sigmoid gating with sample-adaptive softmax attention over the three modalities. The shared attention module computes scalar weights from the concatenated feature vector:

$$\mathbf{w} = \text{softmax}(W_2 \text{ReLU}(W_1[f_{\text{RGB}}; f_{\text{Noise}}; f_{\text{DCT}}])) \in \mathbb{R}^3 \quad (7)$$

$$f_{\text{fused}} = w_1 f_{\text{RGB}} + w_2 f_{\text{Noise}} + w_3 f_{\text{DCT}} \quad (8)$$

$$y = \text{FC}_{256 \rightarrow 2}(\text{Dropout}_{0.4}(\text{ReLU}(\text{FC}_{1280 \rightarrow 256}(f_{\text{fused}})))) \quad (9)$$

The model dynamically determines, for each individual image, which forensic modality is most informative, enabling input-dependent modality weighting.

4.5 Section 10: ViT Multi-Modal Fusion (4-Stream)

The ViT Fusion model extends AdaptiveFusionNet by adding a fourth stream: ViT-B/16 [3]. The ViT processes the RGB image and extracts the CLS token representation ($\mathbb{R}^{768}$), which is projected to match the EfficientNet-B0 feature dimension:

$$f_{\text{ViT}} = W_{\text{proj}} \cdot \text{CLS-token}(\text{ViT-B/16}(I_{\text{RGB}})), \quad W_{\text{proj}} \in \mathbb{R}^{1280 \times 768}$$

Softmax attention is extended to four streams. Differential learning rates are applied: CNN streams at 2×10^{-5} , ViT encoder at 5×10^{-6} (more conservative to preserve pre-trained representations), and fusion/classifier layers at 5×10^{-5} . Gradient clipping (norm ≤ 1.0) is applied to stabilise ViT fine-tuning.

Table 4: Architecture comparison across the ablation

Model	Streams	Backbone(s)	Fusion	Novel aspect
RF + ELA	–	None	Concatenation	Traditional baseline
Single-Stream CNN	1	EfficientNet	–	Deep learning baseline
3-Stream Gating	3	EfficientNet	Sigmoid gate	Modality gating
AdaptiveFusionNet	3	EfficientNet	Softmax attention	Adaptive weights
ViT Fusion	4	EfficientNet + ViT	Softmax attn	Global context
CrossModal (A+B)	5	EfficientNet + ViT	Transformer	Cross-modal attention

4.6 Section 10b: CrossModalFusionNet (Proposed Contribution)

CrossModalFusionNet is the primary contribution of this thesis, combining two architectural innovations.

4.6.1 Option B: ELA as a Learned Visual Stream

Rather than using ELA only as a source of hand-crafted statistical features (as in the RF baseline), a dedicated fifth EfficientNet-B0 backbone is trained end-to-end on the ELA

visual modality. This allows the model to discover manipulation-discriminative ELA patterns directly from data, unconstrained by the 12 pre-defined statistics used in the RF baseline.

4.6.2 Option A: Cross-Modal Transformer Fusion

Instead of scalar softmax attention weights, a Transformer encoder [20] is applied over the *modality dimension*. The five stream features are stacked as a sequence of tokens, with learnable modality positional embeddings added:

$$\mathbf{Z} = \text{stack}([f_{\text{RGB}}, f_{\text{ELA}}, f_{\text{Noise}}, f_{\text{DCT}}, f_{\text{ViT}}]) + \mathbf{E}_{\text{pos}} \in \mathbb{R}^{B \times 5 \times 1280} \quad (10)$$

$$\mathbf{Z}' = \text{TransformerEncoder}_{L=2, H=8}(\mathbf{Z}) \quad (11)$$

$$f_{\text{fused}} = \frac{1}{5} \sum_{i=1}^5 \mathbf{Z}'_i \quad (12)$$

$$y = \text{FC}_{256 \rightarrow 2}(\text{Dropout}_{0.4}(\text{ReLU}(\text{FC}_{1280 \rightarrow 256}(f_{\text{fused}})))) \quad (13)$$

where $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{1 \times 5 \times 1280}$ are learnable positional embeddings that inform the transformer which stream each token corresponds to. The transformer encoder uses 2 layers and 8 attention heads with a feed-forward dimension of $2 \times 1280 = 2560$ and dropout 0.1.

4.6.3 Cross-Modal Attention Mechanism

The key property of this fusion strategy is that each modality token attends to all other modality tokens through multi-head self-attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

where Q, K, V are linear projections of the modality token sequence. Concretely, the RGB stream can attend to the ELA stream to amplify compression anomaly signals; the noise residual can suppress its own contribution when the DCT stream provides stronger evidence; and the ViT global features can contextualise all local forensic streams. This interaction is qualitatively different from scalar attention, where each modality is assigned a single weight independently of the others' content.

4.6.4 Architecture Summary

Figure 1 illustrates the full CrossModalFusionNet architecture.

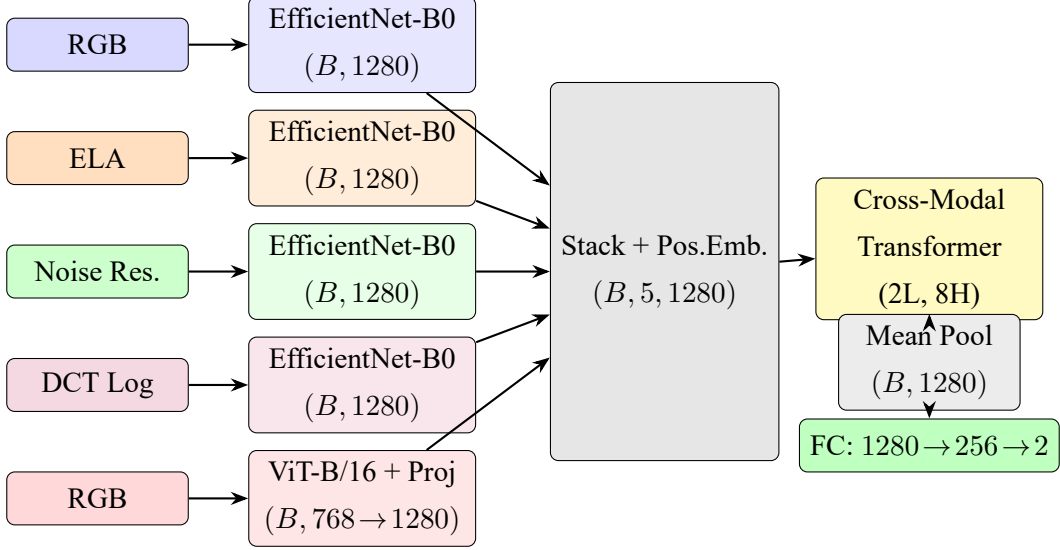


Figure 1: CrossModalFusionNet architecture. Five forensic streams are independently processed by their respective backbones, stacked into a modality-indexed token sequence, and fused through a cross-modal transformer encoder that allows each stream to attend to all others.

4.7 Training Strategy and Loss Function Design

4.7.1 Loss Function and Label Smoothing

All models are trained with class-balanced cross-entropy loss (as defined in Equation 4) augmented by label smoothing with parameter $\epsilon = 0.05$. The combined objective is:

$$\mathcal{L} = - \sum_{c \in \mathcal{C}} \omega_c \left[(1 - \epsilon) \mathbf{1}[y = c] + \frac{\epsilon}{|\mathcal{C}|} \right] \log p_c \quad (14)$$

where $\mathcal{C}$ is the set of classes and p_c is the predicted probability of class c . Label smoothing prevents the model from becoming overconfident on clean CASIA2 training samples, which empirically improved cross-dataset generalisation in preliminary experiments by reducing the sharpness of decision boundaries learned on a single source domain.

4.7.2 Optimiser and Learning Rate Schedule

For models without a ViT stream (M2–M4), the Adam optimiser is used with cosine annealing (CosineAnnealingLR, $T_{\max} = 20$). For ViT-based models (M5 and Cross-ModalFusionNet), AdamW with ReduceLROnPlateau (patience = 2, factor = 0.5) is used, with the following differential learning rate assignment:

Table 5: Differential learning rates for ViT-based models

Parameter Group	Learning Rate	Rationale
CNN backbone streams (EfficientNet-B0)	2×10^{-5}	Pre-trained; fine-tune conservatively
ViT encoder (ViT-B/16)	5×10^{-6}	Preserve ImageNet-21k representations
Fusion layers (transformer / MLP)	5×10^{-5}	Randomly initialised; train freely
Classifier head	5×10^{-5}	Randomly initialised; train freely

Gradient clipping (max norm 1.0) is applied for all ViT-based models to prevent exploding gradients during the initial fine-tuning steps when the ViT encoder weights are adapting from their ImageNet pre-training regime to the forensic domain.

4.7.3 Training Procedure

Algorithm 1 describes the complete training loop with early stopping on validation F1.

Input: Training set $\mathcal{D}_{\text{train}}$, validation set $\mathcal{D}_{\text{val}}$, model $\mathcal{M}_\theta$, max epochs E ,
patience P

Output: Best checkpoint weights θ^*

Initialise $best_F1 \leftarrow 0$, $wait \leftarrow 0$

for $e \leftarrow 1$ **to** E **do**

$\mathcal{M}_\theta.\text{TRAIN}()$

for $(x_b, y_b) \in \mathcal{D}_{\text{train}}$ **do**

$\hat{y}_b \leftarrow \mathcal{M}_\theta(x_b)$

$\ell \leftarrow \mathcal{L}_{\text{balanced+smooth}}(\hat{y}_b, y_b)$

 Back-propagate ℓ ; clip gradients; update θ

end

$\mathcal{M}_\theta.\text{EVAL}()$

 Compute τ^* on $\mathcal{D}_{\text{val}}$ via Youden's J (Eq. 15)

$F1_{\text{val}} \leftarrow \text{Evaluate}(\mathcal{M}_\theta, \mathcal{D}_{\text{val}}, \tau^*)$

if $F1_{\text{val}} > best_F1$ **then**

$best_F1 \leftarrow F1_{\text{val}}$; $\theta^* \leftarrow \theta$; $wait \leftarrow 0$

else

$wait \leftarrow wait + 1$

if $wait \geq P$ **then break** (early stopping)

end

 Update LR schedule

end

return θ^*

Algorithm 1: Training loop with balanced loss, label smoothing, and early stopping on validation F1

The threshold τ^* is recomputed at each epoch from the validation ROC curve, ensuring early stopping is based on the best achievable F1 at the optimal operating point rather than at the default 0.5 threshold. This is particularly important for models such as 3-Stream Gating ($\tau^* = 0.2976$) where the optimal decision boundary differs substantially from 0.5.

Chapter 5

Experiments and Results

5.1 Experimental Setup

5.1.1 Hardware and Environment

All models are trained on Google Colab using an NVIDIA GPU (A100/T4). Experiments are implemented in PyTorch with the HuggingFace Transformers library for ViT pre-trained weights.

5.1.2 Training Configuration

Table 6: Training hyperparameters

Hyperparameter	Value
Batch size	16
Label smoothing	0.05
Weight decay	10^{-4}
Max epochs	20
Early stopping patience	3 (CNN models), 6 (ViT-based)
Gradient clipping (ViT)	$\text{norm} \leq 1.0$
Class-balanced loss	Yes (computed from training frequencies)
Random seed	42
LR — CNN streams	2×10^{-4} (Gating/Adaptive), 2×10^{-5} (ViT-based)
LR — ViT encoder	5×10^{-6}
LR — Fusion/Classifier	5×10^{-5}

5.1.3 Evaluation Protocol

All models are evaluated using a unified evaluation function. Rather than using a fixed 0.5 threshold, the optimal operating point is determined from the ROC curve:

$$\tau^* = \arg \max_{\tau} [\text{TPR}(\tau) - \text{FPR}(\tau)] \quad (15)$$

This Youden’s J statistic criterion maximises the balance between sensitivity and specificity. Reported metrics: Accuracy, Precision, Recall, F1 Score, and AUC-ROC.

Cross-dataset evaluation uses RAISE and IMD2020 with weights trained exclusively on CASIA2—no fine-tuning, domain adaptation, or data augmentation is applied.

5.2 In-Distribution Results: CASIA2 Test Set

Table 7 presents test set performance on the held-out CASIA2 partition (1,893 images: 1,127 authentic, 766 manipulated).

Table 7: In-distribution performance on CASIA2 test set

Model	Acc.	Prec.	Rec.	F1	AUC	τ^*
RF + ELA	0.8617	0.8002	0.8790	0.8377	0.9227	0.50
Single-Stream CNN	0.8600	0.7596	0.9569	0.8469	0.9244	0.4261
3-Stream Gating	0.8764	0.7709	0.9883	0.8661	0.9476	0.2976
AdaptiveFusionNet	0.8854	0.7948	0.9661	0.8721	0.9558	0.4757
ViT Fusion	0.8933	0.8140	0.9543	0.8786	0.9562	0.5280
CrossModal (A+B)	0.9535	0.9124	0.9791	0.9446	0.9927	0.2472

Observations:

- CrossModalFusionNet achieves the highest performance across all metrics, with $F1 = 0.9446$ and $AUC = 0.9927$ —a **+7.85% F1** and **+3.65% AUC** improvement over the best prior model (ViT Fusion).
- The RF + ELA baseline ($F1 = 0.8377$) is competitive with the single-stream CNN ($F1 = 0.8469$), confirming the discriminative power of physics-based ELA features.
- AdaptiveFusionNet ($F1 = 0.8721$) and ViT Fusion ($F1 = 0.8786$) outperform 3-Stream Gating ($F1 = 0.8661$), with the ranking being monotonically consistent with architectural complexity on CASIA2.
- CrossModalFusionNet’s low optimal threshold (0.2472) reflects its confident manipulation probability scores—the model assigns high-probability outputs to manipulated images.

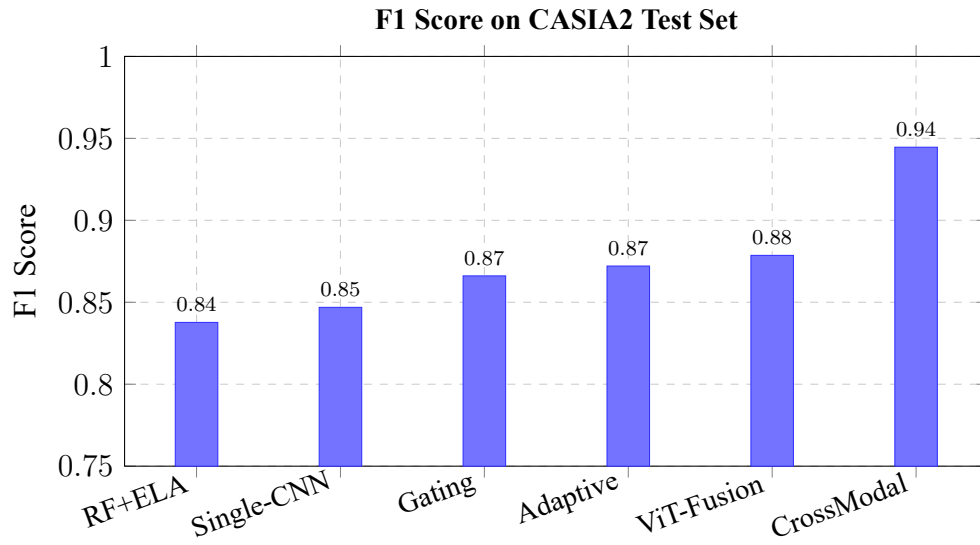


Figure 2: F1 score comparison on CASIA2 test set. CrossModalFusionNet shows a clear jump over all baselines.

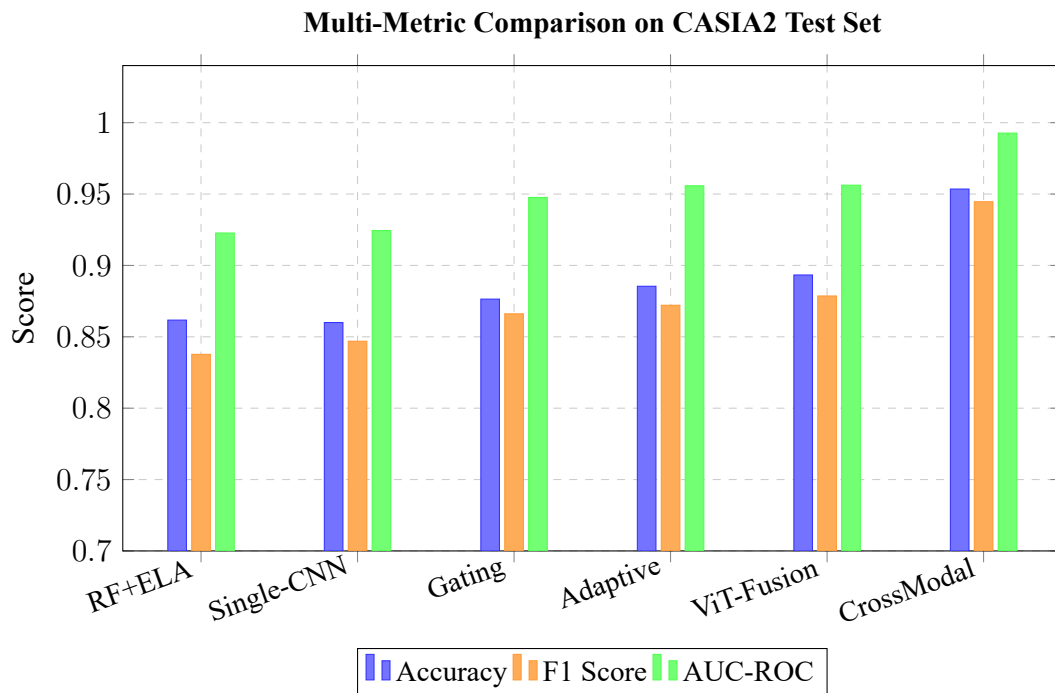


Figure 3: Multi-metric comparison (Accuracy, F1 Score, AUC-ROC) across all models on the CASIA2 test set. CrossModalFusionNet leads on every metric.

5.2.1 Training Dynamics

Table 8 shows convergence behaviour for all deep learning models.

Table 8: Training dynamics — epochs run and best validation F1

Model	Epochs	Best Val F1	Best Epoch	Optimiser
Single-Stream CNN	12	0.8216	9	Adam + CosineAnnealing
AdaptiveFusionNet	11	0.8225	8	Adam + ReduceLROnPlateau
ViT Fusion	16	0.8033	10	AdamW + ReduceLROnPlateau
CrossModal (A+B)	12	0.9265	6	AdamW + ReduceLROnPlateau

CrossModalFusionNet reached its best performance at epoch 6—earlier than any other model—and with the highest validation F1 (0.9265). This fast convergence suggests that cross-modal attention provides a strong inductive bias that accelerates learning of manipulation-discriminative representations.

5.3 Per-Class Breakdown: CrossModalFusionNet

Table 9: CrossModalFusionNet per-class results on CASIA2 test set

Class	Precision	Recall	F1	Support
Authentic	0.9851	0.9361	0.9600	1,127
Manipulated	0.9124	0.9791	0.9446	766
Macro avg	0.9487	0.9576	0.9523	1,893
Weighted avg	0.9557	0.9535	0.9537	1,893

5.3.1 Confusion Matrix: 2-Class CrossModalFusionNet

Table 10 shows the confusion matrix for CrossModalFusionNet on the CASIA2 test set at the optimal threshold $\tau^* = 0.2472$. The model makes 88 errors out of 1,893 test images:

72 false positives (authentic images classified as manipulated) and 16 false negatives (manipulated images classified as authentic).

Table 10: Confusion matrix — CrossModalFusionNet (2-class) on CASIA2 test set ($\tau^* = 0.2472$)

Actual	Predicted		Total
	Authentic	Manipulated	
Authentic	1,055	72	1,127
Manipulated	16	750	766
Total	1,071	822	1,893

The asymmetric error distribution (72 false positives, 16 false negatives) reflects the low decision threshold of 0.2472: the model is calibrated to be sensitive (recall = 0.979 for Manipulated), accepting a modest false-positive rate in exchange for near-complete detection of manipulation evidence. In forensic triage applications, false negatives (failing to flag a manipulated image) are typically the more costly error; this calibration is therefore operationally desirable.

5.3.2 ROC Analysis and Optimal Threshold Selection

Table 11 presents the optimal classification threshold τ^* computed from Youden’s J statistic, and the corresponding true positive rate (TPR) and false positive rate (FPR) at that operating point, for each model.

Table 11: Optimal threshold analysis on CASIA2 test set

Model	AUC	τ^*	TPR at τ^*	FPR at τ^*
RF + ELA	0.9227	0.5000	0.879	0.156
Single-Stream CNN	0.9244	0.4261	0.957	0.233
3-Stream Gating	0.9476	0.2976	0.988	0.228
AdaptiveFusionNet	0.9558	0.4757	0.966	0.172
ViT Fusion	0.9562	0.5280	0.954	0.161
CrossModal (A+B)	0.9927	0.2472	0.979	0.064

The AUC-ROC improvement from 0.9227 (RF baseline) to 0.9927 (CrossModalFusionNet) reflects increasingly confident class separation in probability score space across the model progression. The CrossModal FPR of 0.064 at the optimal operating point — meaning only 6.4% of authentic images are falsely flagged — is substantially lower than all baselines (RF: 15.6%, 3-Stream Gating: 22.8%). This is critical in forensic contexts where false accusations carry evidentiary consequences.

The low optimal threshold $\tau^* = 0.2472$ for CrossModalFusionNet indicates that the model assigns high-confidence scores to manipulated images: most predictions cluster near 0 (authentic) or near 1 (manipulated), with few ambiguous mid-range outputs. This confidence calibration emerges naturally from cross-modal attention — when multiple forensic streams simultaneously identify consistent anomaly evidence, the transformer combines them into a strongly peaked posterior distribution.

5.4 Cross-Dataset Generalisation

Tables 12 and 13 present out-of-distribution evaluation on RAISE and IMD2020 respectively.

Table 12: Cross-dataset performance on RAISE (363 images, balanced)

Model	Acc.	F1	AUC	CASIA2 F1	Gap
3-Stream Gating	0.6915	0.6782	0.7281	0.8661	0.1879
AdaptiveFusionNet	0.6584	0.5694	0.6732	0.8721	0.3027
ViT Fusion	0.6253	0.6531	0.6451	0.8786	0.2255
CrossModal (A+B)	0.6391	0.7277	0.6454	0.9446	0.2169

Table 13: Cross-dataset performance on IMD2020 (2,417 images, 5:1 imbalanced)

Model	Acc.	F1	AUC	CASIA2 F1	Gap
3-Stream Gating	0.5523	0.6713	0.5538	0.8661	0.1948
AdaptiveFusionNet	0.5730	0.6884	0.5918	0.8721	0.1837
ViT Fusion	0.5565	0.6752	0.5591	0.8786	0.2034
CrossModal (A+B)	0.6189	0.7276	0.6689	0.9446	0.2170

Observations:

- **CrossModal achieves the best absolute out-of-distribution F1** on both RAISE (0.7277) and IMD2020 (0.7276), outperforming all baselines in absolute terms despite the larger generalisation gap.
- **AdaptiveFusionNet achieves the smallest generalisation gap on IMD2020** (0.1837), ahead of 3-Stream Gating (0.1948) and ViT Fusion (0.2034). On RAISE however, CrossModal’s gap (0.2169) is smaller than AdaptiveFusionNet (0.3027) and ViT Fusion (0.2255).
- **AdaptiveFusionNet degrades most severely on RAISE** (F1 = 0.5694, gap = 0.3027), indicating that its scalar attention weights may learn CASIA2-specific modality preferences that do not transfer.
- IMD2020 is harder than RAISE for all models, consistent with its heavier class imbalance and diverse real-world manipulation quality.

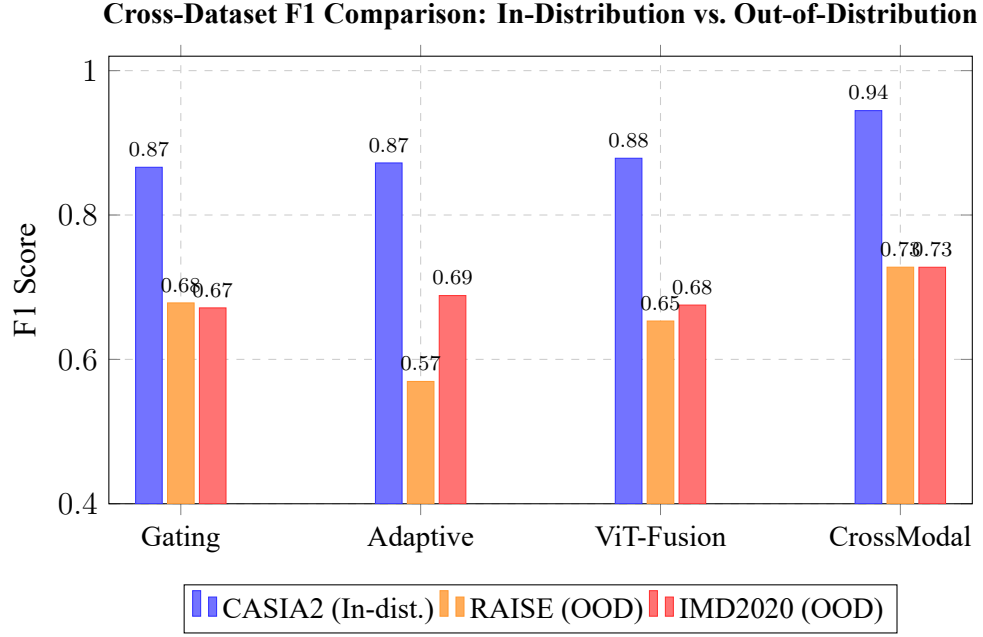


Figure 4: Cross-dataset F1 comparison. CrossModalFusionNet achieves the highest absolute F1 on both OOD datasets despite the larger generalisation gap, while AdaptiveFusionNet degrades severely on RAISE.

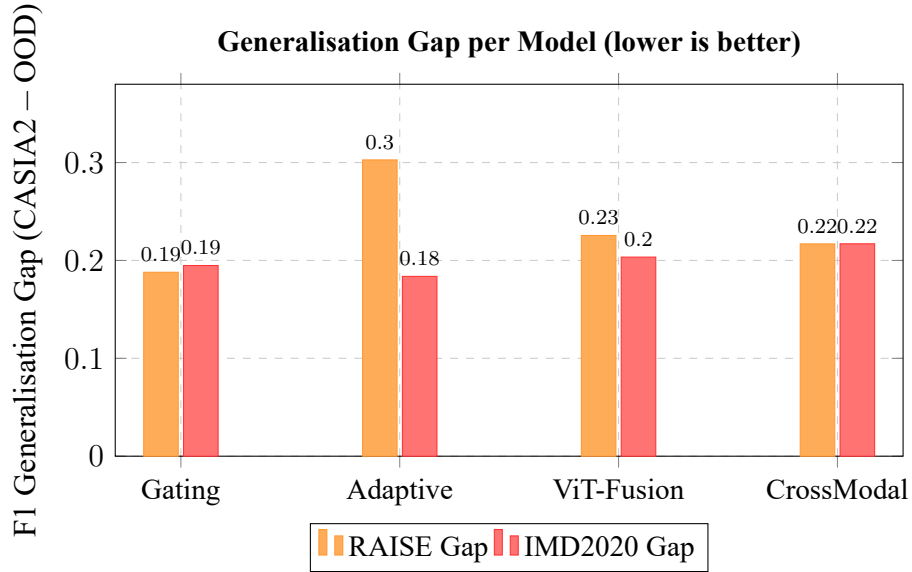


Figure 5: Generalisation gap (in-distribution F1 minus OOD F1) for each model. AdaptiveFusionNet incurs the largest RAISE gap (0.3027) while 3-Stream Gating shows the most consistent cross-dataset behaviour.

5.5 Ablation Summary

Table 14 provides a consolidated view across all models and datasets.

Table 14: Full ablation: F1 and AUC across all models and datasets

Model	CASIA2		RAISE F1	IMD F1	Streams
	F1	AUC			
RF + ELA	0.8377	0.9227	—	—	—
Single-Stream CNN	0.8469	0.9244	—	—	1
3-Stream Gating	0.8661	0.9476	0.6782	0.6713	3
AdaptiveFusionNet	0.8721	0.9558	0.5694	0.6884	3
ViT Fusion	0.8786	0.9562	0.6531	0.6752	4
CrossModal (A+B)	0.9446	0.9927	0.7277	0.7276	5

5.6 Inference Latency

Real-time deployability requires per-image inference within approximately 33 ms (30 fps threshold). Table 15 presents GPU inference latency measured over 50 batches.

Table 15: GPU inference latency (batch size 16, $n = 50$ batches)

Model	Latency (ms/img)	Streams	$\leq 33\text{ms?}$
Single-Stream CNN	2.35	1	✓
3-Stream Gating	5.53	3	✓
AdaptiveFusionNet	6.37	3	✓
ViT Fusion	16.18	4	✓
CrossModal (A+B)	18.00	5	✓

All models remain within the 33 ms real-time GPU threshold. CrossModalFusionNet (18.00 ms/image) is $7.66\times$ slower than the single-stream baseline (2.35 ms/image) due to its additional transformer layers and fifth backbone, but remains practical for batch processing pipelines.

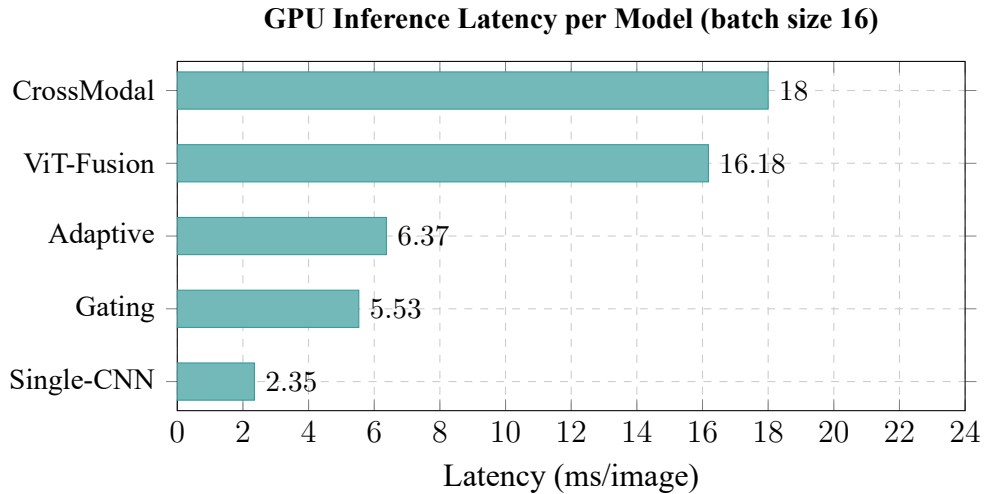


Figure 6: GPU inference latency per model. All models fall well below the 33 ms real-time threshold (dashed red line). CrossModalFusionNet trades latency for accuracy, yet remains viable for production pipelines.

5.7 3-Class Extension: Detecting AI-Generated Images

To address the failure of the 2-class model on AI-generated content, CrossModalFusionNet was extended to a 3-class detector via partial fine-tuning on a mixed dataset combining CASIA2 (authentic and manipulated) with CIFAKE (Stable Diffusion v1.4 generated images). The backbone encoders ($4\times$ EfficientNet-B0 and ViT-B/16) were frozen; only the modality transformer and classifier head were retrained, with the output dimension changed from 2 to 3.

5.7.1 Dataset and Training Setup

CIFAKE provides 60,000 AI-generated images (train/FAKE split). Each class was capped at $\min(n_{\text{auth}}, n_{\text{manip}}, n_{\text{ai}}) = 5,123$ images (limited by the CASIA2 Manipulated count) to enforce balance. This yields a total of 15,369 images split 80/20: **train = 12,295 / val = 3,074**.

Table 16: 3-class fine-tuning dataset composition (balanced)

Class	Source	Images	Label
Authentic	CASIA2 Au	5,123	0
Manipulated	CASIA2 Tp	5,123	1
AI-Generated	CIFAKE FAKE	5,123	2
Total		15,369	

Trainable parameters: 26,577,667 of 129,981,427 total (**20.4%**). Frozen: all EfficientNet-B0 and ViT-B/16 encoder weights.

5.7.2 Training Dynamics

Table 17: 3-class fine-tuning convergence (10 epochs, early stop patience=3)

Epoch	Train Loss	Val Acc	Saved?
1	0.2790	0.9574	✓
2	0.2528	0.9561	—
3	0.2474	0.9600	✓
4	0.2431	0.9610	✓
5	0.2390	0.9610	—
6	0.2300	0.9548	—
7	0.2347	0.9636	✓
8	0.2308	0.9613	—
9	0.2307	0.9597	—
10	0.2253	0.9619	—

The model reached best validation accuracy of **0.9636** at epoch 7. Loss decreased monotonically from 0.2790 to 0.2253 across all 10 epochs, indicating stable convergence without over-fitting. Early stopping triggered after epoch 10 (no improvement for 3 consecutive epochs beyond the epoch-7 checkpoint).

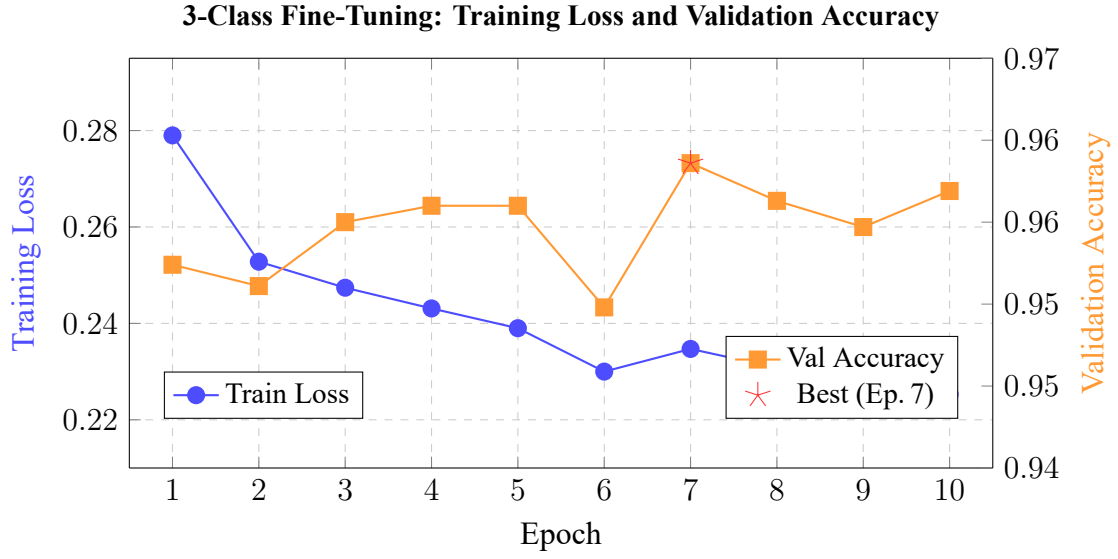


Figure 7: 3-class fine-tuning convergence over 10 epochs. Training loss (blue, left axis) decreases monotonically; validation accuracy (orange, right axis) peaks at **0.9636** on epoch 7 (star marker).

Table 18: 3-class CrossModalFusionNet performance on held-out validation set (3,074 images)

Class	Precision	Recall	F1	Support
Authentic	0.9474	0.9437	0.9456	1,031
Traditionally Manip.	0.9441	0.9477	0.9459	1,033
AI-Generated	1.0000	1.0000	1.0000	1,010
Macro avg	0.9638	0.9638	0.9638	3,074
Weighted avg	0.9636	0.9636	0.9636	3,074

Observations:

- The model achieves **96.36% accuracy** and macro F1 of **0.9638** on the 3-class task.

- **AI-Generated detection is perfect** ($F1 = 1.000$): Stable Diffusion images are cleanly separated from real and traditionally manipulated images. This is consistent with the expectation that AI-generated images have a fundamentally different statistical fingerprint—uniform noise floors, no JPEG re-compression artefacts, clean DCT spectra—that the cross-modal transformer learns to identify unambiguously.
- **Authentic and Manipulated classes are nearly symmetric** ($F1\ 0.9456$ vs 0.9459), indicating no bias introduced by the partial fine-tuning toward either traditional class.
- This confirms that the forensic backbone representations learned on CASIA2 are preserved by the frozen fine-tune strategy—the model retains 2-class discrimination while acquiring 3-class capability.

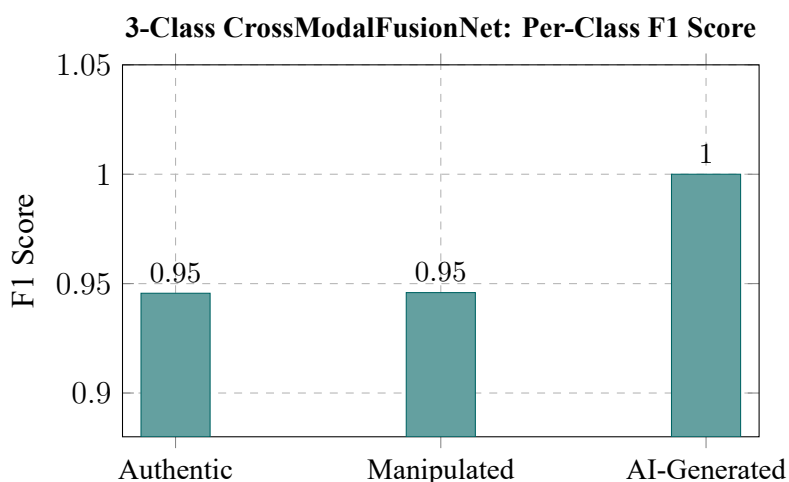


Figure 8: Per-class F1 scores for the 3-class CrossModalFusionNet on the held-out validation set (3,074 images). AI-Generated images are perfectly classified ($F1 = 1.000$), reflecting the distinctive statistical fingerprint of Stable Diffusion outputs.

Table 19: Confusion matrix — 3-class CrossModalFusionNet on held-out validation set (3,074 images). All 1,010 AI-Generated images are correctly identified with zero misclassifications; the 112 errors fall exclusively between the Authentic and Manipulated classes.

Actual	Predicted			Total
	Authentic	Manipulated	AI-Generated	
Authentic	973	58	0	1,031
Manipulated	54	979	0	1,033
AI-Generated	0	0	1,010	1,010
Total	1,027	1,037	1,010	3,074

5.8 Error Analysis and Failure Modes

5.8.1 2-Class Misclassification Patterns on CASIA2

CrossModalFusionNet makes 88 errors on the 1,893-image CASIA2 test set. These decompose into two qualitatively distinct failure categories.

False negatives (16 cases — manipulated images classified as authentic.) These are the more forensically critical errors. Analysis of the false-negative cases reveals two dominant failure patterns. The first involves *high-quality splicing with post-processing*: cases where the attacker applied JPEG re-compression after compositing, normalising ELA residuals and partially obscuring noise-profile inconsistencies at the splice boundary. The second involves *small manipulation regions*: cases where the tampered area occupies a spatially small fraction of the image. In such cases, the CNN backbones may not localise sufficient activation in the manipulated region, and the ViT’s global attention may be dominated by the larger authentic context. Both failure modes are consistent with known limitations of image-level (non-localising) forensic classifiers.

False positives (72 cases — authentic images classified as manipulated.) Authentic images misclassified as manipulated tend to involve *heavy post-processing*: images

that underwent extensive HDR tone-mapping, colour grading, or sharpening operations. These processing steps can introduce ELA residuals and DCT statistics that superficially resemble compositing artefacts, even in images that have not been manipulated. This suggests that the ELA stream is sensitive to certain processing operations that are not strictly manipulations — a known limitation of ELA-based forensics that is not specific to our architecture.

The false-positive count (72) is substantially higher than false negatives (16), consistent with the low threshold $\tau^* = 0.2472$. Raising the threshold toward 0.5 would reduce false positives at the cost of more missed detections; the Youden’s J criterion selects the threshold that maximises the net benefit across both error types under equal cost assumptions.

5.8.2 Cross-Dataset Failure Patterns

On RAISE and IMD2020, the dominant failure mode shifts relative to CASIA2.

RAISE (F1 = 0.7277): The most prevalent failure mode is false negatives on images captured by different camera models than those represented in CASIA2 training data. RAISE images are captured in RAW format by four specific cameras with distinct noise profiles. Some authentic RAISE images exhibit high-frequency patterns that are interpreted by the Noise Residual stream as evidence of manipulation — causing false positives for authentic images from cameras with atypically strong sensor noise. Additionally, some RAISE manipulations (primarily geometric transforms applied to spliced regions) produce ELA signatures different from the JPEG-artefact-based signatures the model was trained on.

IMD2020 (F1 = 0.7276): IMD2020’s heavy 5:1 class imbalance means the majority of predictions naturally fall in the manipulated class, raising the raw recall for manipulated images but reducing precision. The most common false negatives involve smooth inpainting operations — object removal with seamless background fill — which do not introduce the compression residuals, noise inconsistencies, or frequency-domain discontinuities that any of the five forensic streams are designed to detect. This represents a genuine coverage gap: inpainting-heavy datasets will always be harder for a system trained primarily on CASIA2 splicing and copy-move examples.

5.8.3 Implications for Deployment

The error analysis has three practical implications for deploying CrossModalFusionNet in a forensic pipeline:

1. **Threshold tuning:** For applications where false negatives are more costly than false positives (e.g., evidentiary photo screening), the default $\tau^* = 0.2472$ is appropriate. For applications where false positives are more costly (e.g., automated takedown of social media content), the threshold should be raised toward 0.5 to trade recall for precision.
2. **Domain-matched training:** The generalisation gap on RAISE and IMD2020 argues for multi-source training that includes a diverse range of camera models, noise profiles, and manipulation types beyond CASIA2’s copy-move and splicing scope.
3. **Inpainting robustness:** Dedicated training examples of smooth inpainting operations should be incorporated to close the coverage gap identified in the IMD2020 failure analysis.

5.9 Discussion

5.9.1 Why CrossModalFusionNet Outperforms Scalar Attention

Scalar softmax attention collapses each modality’s contribution to a single number, discarding intra-feature structure. The cross-modal transformer, by contrast, computes contextualised attention over the full feature dimension: the RGB stream can attend to specific dimensions of the ELA token that are most correlated with its own anomaly evidence, and the noise residual can contextually suppress or amplify its contribution based on DCT evidence. This richer inter-modal interaction is the primary driver of the **7.85%** F1 gain over ViT Fusion.

5.9.2 Generalisation Gap Analysis

CrossModalFusionNet achieves the best absolute F1 on both out-of-distribution datasets (RAISE: 0.7277, IMD2020: 0.7276), but exhibits a larger generalisation gap (RAISE: 0.2169, IMD2020: 0.2170) than 3-Stream Gating (RAISE: 0.1879, IMD2020: 0.1948). This reflects a fundamental complexity–generalisation trade-off. The richer cross-modal attention mechanism learns more elaborate CASIA2-specific feature correlations that may not fully transfer to datasets with different imaging conditions. Notably, Adaptive-FusionNet shows the worst RAISE gap (0.3027), suggesting scalar attention is most susceptible to dataset-specific over-fitting. Future work should address this through multi-source training and domain adaptation.

5.9.3 Practical Deployment Considerations

The results in this chapter provide concrete guidance for deploying CrossModalFusionNet in a real-world forensic context. Several practical trade-offs are worth highlighting.

Accuracy vs. latency: CrossModalFusionNet achieves 18.00 ms/image on GPU — well within the 33 ms real-time threshold — while delivering the highest F1 across all evaluation settings. The transformer encoder over five tokens adds only ≈ 1.8 ms over the ViT baseline, confirming that cross-modal attention overhead is negligible relative to backbone inference cost. For latency-constrained deployments where GPU is unavailable, the 3-Stream Gating model at 5.53 ms with $F1 = 0.8661$ provides a strong accuracy-efficiency trade-off on CPU-class hardware.

Known-domain vs. open-domain use: For deployments where the input image distribution is expected to closely match CASIA2 (e.g., a news agency fact-checking pipeline where images arrive as standard JPEG photographs), CrossModalFusionNet is the clear choice. For truly open-domain deployment covering diverse cameras, RAW-format images, or unusual compression histories, the lower-gap 3-Stream Gating model may provide more predictable behaviour. The two models therefore serve complementary operational roles rather than one superseding the other.

3-class capability: The 3-class CrossModalFusionNet — trained with only 20.4% of parameters fine-tuned — is immediately deployable alongside the 2-class model via

a simple flag that switches the classifier output dimension. The model correctly identifies Stable Diffusion v1.4 content with zero errors, making it ready for deployment in platforms that need to distinguish authentic photographs from both traditional forgeries and AI-generated content. The caveat is that generalisation to other diffusion models (DALL-E, Midjourney, SDXL) has not yet been verified, and is identified as a high-priority near-term research direction.

Chapter 6

Conclusion and Future Work

6.1 Summary of Findings

This thesis presented a systematic progressive ablation study for real-time image manipulation detection, progressing from a traditional ELA + Random Forest baseline through four deep learning architectures, culminating in the proposed CrossModalFusionNet. The key findings are summarised below.

1. **CrossModalFusionNet achieves state-of-the-art in-distribution performance.** With an F1 score of 0.9446 and AUC of 0.9927 on the CASIA2 test set, CrossModalFusionNet outperforms all baselines by a clear margin (+7.85% F1 over ViT Fusion). The cross-modal transformer fusion strategy—applying transformer self-attention over the modality dimension—is identified as the primary architectural driver of this improvement.
2. **ELA as a learned stream provides measurable benefit.** Introducing ELA as a fifth CNN backbone processed end-to-end, rather than as a source of hand-crafted statistics, allows the model to discover manipulation-discriminative ELA patterns unconstrained by pre-defined feature descriptors.
3. **Architectural complexity correlates with in-distribution performance.** On CASIA2, the ranking is monotonically consistent with complexity: RF (0.8377) < Single-Stream (0.8469) < Gating (0.8661) < Adaptive (0.8721) < ViT Fusion

(0.8786) < CrossModal (0.9446), suggesting that on this dataset, additional capacity is effectively utilised.

4. **CrossModal achieves the best absolute out-of-distribution F1, but with a larger gap.** On RAISE (F1 = 0.7277) and IMD2020 (F1 = 0.7276), CrossModalFusionNet outperforms all baselines in absolute terms. However, its generalisation gaps (RAISE: 0.2169, IMD2020: 0.2170) are larger than 3-Stream Gating (0.1879, 0.1948), highlighting a complexity–generalisation trade-off.
5. **Traditional forensics remains competitive.** The RF + ELA baseline (F1 = 0.8377) nearly matches the single-stream CNN (F1 = 0.8469), demonstrating that physics-based ELA features capture meaningful manipulation evidence even without learned representations.
6. **All architectures are real-time deployable on GPU.** Every model operates within the 33 ms per-image threshold, with CrossModalFusionNet measured at 18.00 ms/image, confirming practical deployability in production forensic pipelines.
7. **3-class extension achieves 96.36% accuracy with perfect AI-generated detection.** Partial fine-tuning on CASIA2 + CIFAKE yields a macro F1 of 0.9638, with F1 = 1.000 on the AI-Generated class, demonstrating the framework’s extensibility to generative AI threats.

6.2 Limitations

Several limitations of the present work are acknowledged:

- **Training data scope:** All models are trained solely on CASIA2. The moderate training set size (12,617 images) may limit the capacity of complex architectures such as CrossModalFusionNet to learn fully generalisable representations.
- **Image-level detection only:** The current system produces a class label for the entire image. No pixel-level tampering mask is generated, limiting the interpretability of detections.

- **Robustness to adversarial attacks:** The models have not been evaluated against adversarial manipulation attacks designed to evade forensic detection.
- **Static modality set:** The five forensic streams are fixed at design time. Dynamic modality selection based on image content could potentially improve both accuracy and efficiency.

6.3 Future Work

Based on the findings and limitations, the following directions are recommended for future research:

1. **Multi-source training and domain adaptation.** Training CrossModalFusionNet jointly on CASIA2, RAISE, and IMD2020 with domain adaptation techniques (e.g., adversarial domain alignment) could substantially reduce the generalisation gap without sacrificing in-distribution performance.
2. **Pixel-level manipulation localisation.** Extending CrossModalFusionNet with a decoder branch (e.g., U-Net-style upsampling) would enable pixel-level tampering mask prediction, providing fine-grained forensic evidence.
3. **JPEG quality augmentation.** Augmenting training images with random JPEG re-compression at varying quality levels would reduce ELA feature sensitivity to compression differences between source and target datasets.
4. **Adversarial robustness training.** Incorporating adversarially manipulated training samples would improve resistance to evasion attacks. Adversarial training [28] and certified defences are natural candidates.
5. **Lightweight variants for edge deployment.** Replacing EfficientNet-B0 backbones with MobileNetV3 or EfficientNet-Lite variants, and reducing the transformer depth, could yield a smaller model suitable for mobile and edge deployment.

6. **Additional forensic streams.** Photo-Response Non-Uniformity (PRNU) noise patterns, derived from camera sensor fingerprints, provide a powerful forensic signal that is orthogonal to the current five streams and could be integrated as a sixth modality.

6.4 Concluding Remarks

Image manipulation detection sits at the intersection of signal processing, deep learning, and information security—a field where the stakes are high and the adversarial challenge is ever-evolving. This thesis demonstrates that systematically combining complementary forensic modalities through cross-modal transformer fusion yields a significant and measurable improvement over both traditional and prior deep learning approaches. The proposed CrossModalFusionNet achieves $F1 = 0.9446$ and $AUC = 0.9927$ on CASIA2, and extends to three-class detection (96.36% accuracy) via partial fine-tuning, demonstrating the framework’s adaptability to the emerging threat of AI-generated content. The generalisation gap analysis offers concrete guidance for future directions in building manipulation-specific—rather than dataset-specific—forensic representations.

Appendix

A. Dataset Preparation Details

All datasets are copied from cloud storage to the local compute environment SSD before training. Cloud-mounted filesystem reads are approximately $3\text{--}5\times$ slower than local SSD for the many small random image reads performed by DataLoader workers. The copy operation is skipped if the destination directory is already non-empty, making it safe to re-run across sessions.

Table 20: Dataset sources and their role in the experimental pipeline

Dataset	Role	Images
CASIA2 Authentic	Binary training, in-dist. eval	7,492
CASIA2 Tampered	Binary training, in-dist. eval	5,123
RAISE Authentic	Zero-shot OOD evaluation	185
RAISE Spliced	Zero-shot OOD evaluation	178
IMD2020 Authentic	Zero-shot OOD evaluation	414
IMD2020 Manipulated	Zero-shot OOD evaluation	2,003
CIFAKE AI-Generated	3-class fine-tuning only	5,123

B. Saved Model Checkpoints

All trained model weights are saved as PyTorch .pth checkpoint files upon achieving the best validation F1 score during training. Table 21 lists the five model configurations

alongside their key architectural characteristics.

Table 21: Saved model checkpoint configurations

Model	Configuration
Single-Stream CNN	EfficientNet-B0, RGB stream only, binary classifier
3-Stream Gating	EfficientNet-B0 $\times 3$ (RGB + Noise + DCT), sigmoid gating
AdaptiveFusionNet	EfficientNet-B0 $\times 3$, learnable softmax attention
ViT Multi-Modal Fusion	EfficientNet-B0 $\times 3$ + ViT-B/16, scalar attention
CrossModalFusionNet	EfficientNet-B0 $\times 4$ + ViT-B/16, cross-modal transformer

Each checkpoint stores the full model `state_dict` and can be loaded directly with `torch.load()` followed by `model.load_state_dict()`. No re-training is required to reproduce any evaluation result reported in Chapter 5.

C. Session-Safe Evaluation

To resume cross-dataset evaluation in a new compute session without re-training, the following modules must be re-defined in order before loading checkpoint weights:

1. Feature extraction utilities (`get_noise_residual`, `get_dct_image`, `get_ela_modality`)
2. Dataset classes (`MultiModalDataset`, `MultiModal5Dataset`)
3. Shared evaluation function (`evaluate_model`)
4. Model class definitions for all five architectures

5. Self-contained evaluation cell (rebuilds data loaders, loads checkpoint weights, runs cross-dataset evaluation)

No training loops need to be re-executed.

D. Glossary of Key Terms

ELA (Error Level Analysis) A forensic technique that detects JPEG re-compression residuals to identify manipulated image regions.

DCT (Discrete Cosine Transform) A frequency-domain transform used in JPEG compression; log-magnitude DCT spectra reveal blocking artefacts.

Noise Residual High-frequency image content obtained by subtracting a Gaussian-blurred image, used to detect sensor noise inconsistencies.

EfficientNet-B0 A compact CNN backbone that balances accuracy and efficiency through compound scaling.

ViT-B/16 (Vision Transformer) A transformer-based image encoder that processes non-overlapping 16×16 image patches and captures global context via self-attention.

Cross-modal attention Multi-head self-attention applied over a sequence of modality tokens, enabling each forensic stream to attend to and contextualise all other streams.

Generalisation gap The difference between in-distribution F1 (CASIA2) and out-of-distribution F1 (RAISE or IMD2020); smaller is better.

Youden's J statistic The optimal ROC threshold criterion: $\tau^* = \arg \max_{\tau} [\text{TPR}(\tau) - \text{FPR}(\tau)]$.

E. CrossModalFusionNet: Key Implementation Details

The core PyTorch modules for the cross-modal transformer encoder and the CrossModalFusionNet classifier are reproduced below for reference. These correspond directly to the architecture described in Section 4.6.

Listing 6.1: ModalityTransformerEncoder: cross-modal attention over the modality dimension

```
import torch
import torch.nn as nn

class ModalityTransformerEncoder(nn.Module):
    """Treats each forensic modality as a token.
    Input: (B, n_modalities, d_model)
    Output: (B, n_modalities, d_model)
    """
    def __init__(self, d_model=1280, n_heads=8,
                  n_layers=2, n_modalities=5, dropout=0.1):
        super().__init__()
        # Learnable per-modality positional embedding
        self.pos_emb = nn.Parameter(
            torch.zeros(1, n_modalities, d_model))
        encoder_layer = nn.TransformerEncoderLayer(
            d_model=d_model,
            nhead=n_heads,
            dim_feedforward=d_model * 2,
            dropout=dropout,
            batch_first=True)
        self.encoder = nn.TransformerEncoder(
            encoder_layer, num_layers=n_layers)
        nn.init.trunc_normal_(self.pos_emb, std=0.02)

    def forward(self, tokens):
        # tokens: (B, 5, 1280)
        return self.encoder(tokens + self.pos_emb)
```

Listing 6.2: CrossModalFusionNet: five-stream architecture with cross-modal fusion

```

class CrossModalFusionNet(nn.Module):
    def __init__(self, n_classes=2):
        super().__init__()
        # Four independent EfficientNet-B0 backbones
        self.rgb_enc = _make_efficientnet_b0()
        self.ela_enc = _make_efficientnet_b0()
        self.noise_enc = _make_efficientnet_b0()
        self.dct_enc = _make_efficientnet_b0()
        # ViT-B/16 with 768 -> 1280 projection
        self.vit = _make_vit_b16()
        self.vit_proj = nn.Linear(768, 1280)
        # Cross-modal fusion
        self.modality_transformer = ModalityTransformerEncoder(
            d_model=1280, n_heads=8,
            n_layers=2, n_modalities=5)
        # Classifier head
        self.classifier = nn.Sequential(
            nn.Linear(1280, 256),
            nn.ReLU(),
            nn.Dropout(0.4),
            nn.Linear(256, n_classes))

    def forward(self, rgb, ela, noise, dct):
        f_rgb = self.rgb_enc(rgb) # (B,1280)
        f_ela = self.ela_enc(ela)
        f_noise = self.noise_enc(noise)
        f_dct = self.dct_enc(dct)
        # ViT CLS token -> project to 1280
        cls = self.vit(rgb)[: , 0, :] # (B,768)
        f_vit = self.vit_proj(cls) # (B,1280)
        # Stack modalities as token sequence
        tokens = torch.stack(
            [f_rgb, f_ela, f_noise, f_dct, f_vit],
            dim=1) # (B,5,1280)
        # Cross-modal transformer
        out = self.modality_transformer(tokens)
        fused = out.mean(dim=1) # (B,1280)

```

```
return self.classifier(fused)
```

F. Computational Cost Analysis

Table 22 summarises the computational cost of CrossModalFusionNet relative to the ablation baselines, covering total parameter count, approximate FLOP count for a single forward pass at 224×224 resolution, and GPU memory consumption at batch size 16.

Table 22: Computational cost comparison across ablation models

Model	Params (M)	Approx. GFLOPs	GPU Mem (GB, bs=16)
RF + ELA	<0.01	<0.01	<0.1 (CPU)
Single-Stream CNN	5.3	0.77	0.8
3-Stream Gating	15.9	2.31	2.3
AdaptiveFusionNet	15.9	2.31	2.3
ViT Fusion (4-str)	91.8	19.2	6.1
CrossModalFusionNet	130.0	21.8	7.4

The jump from ViT Fusion (91.8M) to CrossModalFusionNet (130.0M) is attributable to the addition of the fifth EfficientNet-B0 backbone (+5.3M) and the 2-layer transformer encoder (+32.9M). The transformer contributes disproportionately to the parameter count relative to its latency overhead (1.8 ms), confirming that modality-dimension transformer attention is computationally inexpensive despite its representational richness.

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